



Building a Process Meta Model Extended for Cycles

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ABSTRACT

Process modeling is an important and delicate step, and it is among reasons that cause failure in the implementation or the improvement of any type of system. Indeed, process modeling is based on the inter-relationship between the processes that make up the system. There are various standards of process modeling; each one presents several aspects and concepts in relation with the process. While taking into account the specific characteristics of each case of modeling, this diversity makes the choice of the model very complicated. In this paper, we use the concepts of the MDE approach to create our process meta model extended for cycles. We generate this meta model out of several transformations of the Process Structure View meta model extended for lifecycles and the concept view meta model, using the Seven Views approach. The meta model proposed will facilitate the task to the process managers in terms of collecting requirements; proposals and solutions of improvement, through a repository containing all the information collected at the theoretical modeling, and updated during each cycle. The objective of this paper is to propose a generic process meta model, extended for cycles, that considers the requirements dimension, using Seven Views approach. This meta model will be valid for all type of information systems and for all type of cycles (lifecycles and improvement cycles).

CCS CONCEPTS

- Applied computing ~ Business process modeling

KEYWORDS

Lifecycle, Improvement cycle, Business Process modeling, Requirements, UML, Seven Views

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ArabWIC 2019, March 7–9, 2019, Rabat, Morocco
© 2019 Association for Computing Machinery.
ACM ISBN 978-1-4503-6089-0/19/03...\$15.00
<https://doi.org/10.1145/3333165.3333182>

1 Introduction

Process modeling is an important step in the lifecycle and the improvement cycle of a process, it allows a description of the process and its different activities, aiming a good performance within the global information system. There are several standards of process modeling, and we can note that the existing models in the literature represent different points of view of the process [1]. These models are expressed in several ways: some are represented as UML diagrams (SPEM [2], OPF [3] and ISO 24744 [4]), others with flowcharts, while some are still described textually [5][6][7] or from examples. It is therefore difficult to compare, evaluate and select the most relevant model in accordance to the needs of each organization. In addition, in the set of process modeling standards, BPMN [8] is becoming the standard of process modeling. The diagrams proposed by BPMN allow capturing perfectly the behavioral aspect and a view of the instantiation of a process. However, it does not support multiple aspects such as the process requirements [9]. In this paper, we are interested in the process modeling, either in the implementation or in the improvement phase of the process. We emphasize that the definition of the requirements related to the process is very important for the good realization of the model. For this reason, we propose to extend the meta model process proposed in the framework of the Seven Views approach [10], using UML diagrams, to the extent that this approach provides a view that formalize the requirements.

The Seven Views approach proposes to offer an undeniable contribution to the process modeling, by defining a simple mechanism for its integration and taking into account several aspects. Supported by the British Computer Society [11], it is among the most comprehensive methods to describe processes. As its name indicates, this approach offers seven views considered essential to fully and completely model a process [10][11], these views are: requirements view, process structure view, process contents view, process behavior view, information view, stakeholders view and process instance view.

In this paper, we will use the concepts of the MDE approach to create our process meta model extended for cycles. We will generate this meta model from several transformations of Process Structure View extended for lifecycles and the Concept View.

The meta model proposed will facilitate the task in terms of collection of requirements, risks & opportunities and

improvement proposal, updating a repository that contains information collected in the theoretical modeling phase.

The extension of the process meta model for the cycles will provide a global view of the process and its different components, as well as its implementation and improvement, which will allow the requirement's formalization and registration in repositories since the process theoretical modeling phase. These repositories will ensure a gain at the level of the process improvement (via the improvement cycle), as they allow a traceability of requirements.

We note that the meta model proposed through this paper concerns each process separately, and does not concern the whole system.

This paper is organized in seven sections: after a general introduction, the second section presents the methodology used to construct the proposed meta model, the third and fourth sections describe in detail the two steps of the methodology. The fifth section outlines a discussion accompanied by a case study, and finally the sixth section presents the related works. We finish with a conclusion and perspectives.

2 Related work

Although there are multiple Process modeling Standards (BPMN [8], UML, EPC, IDEFx and RAD), there is almost no extension of any of them that include or introduce the concept of cycle (Lifecycle or improvement cycle), except an extension proposed by [10], in the framework of Seven Views approach, that extend Process Structure View for Lifecycles. This proposition links the meta class Process and the meta class lifecycle through the meta class iteration. Unless, it represents only one specific case of cycles (lifecycle)

Moreover, there is a lot of extensions which introduce several concepts other than cycles. For example, the BPMN has been extended for several aspects, to describe different process characteristics across distinct domains [12]. These include, but are not limited to: quality management [13][14], performance measurement [15][16], E-health [17], security [18], resources [19], process tailoring [20], Internal control [21] and time [22], Wireless Sensor Networks (WSNs) [23] [24], process event streams from the IoT [25] ... but none of these extensions introduced the cycles.

There are several extensions of the UML language, such as SysML, which is an extension of UML dedicated to complex systems [26], extensions related to security and information risks, namely: SecureUML [27], UMLSec [28] and abuse cases [29]. A. Rodriguez, E. Fernandez-Medina, and M. Piattini proposed in [30] an extension of UML for the modeling of security requirements in business processes.

EPC was extended for many topics, on several occasions, following different points of view; such as risks [31], [32] and [33] where the authors conduct a refinement to the modeling of risks. One of the popular variations in EPC is the configurable EPC (C-EPC). However, there are no specific extensions for lifecycle or improvement cycles.

Concerning the IDEFx languages and RAD, we did not find any extension for lifecycles or improvement cycles.

Our contribution consists of including cycle (Lifecycle or improvement cycle) in the modeling phase, using the Concept View of the Seven Views approach, in order to prevent time waste in collecting process information (risks & opportunities, requirements...).

3 Methodology of Building the Meta Model

In this article, we use the concepts of the MDE approach for the construction of the proposed meta model.

Model Driven Engineering (MDE) is a part of the evolution of the development of computer systems techniques, in order to control their complexity with a more abstract concern than the conventional programming [34]. By relying on generative approaches, it is to generate all or a part of an application from models [35].

In fact, this approach is based on the intensive use of models as well as on the concept of transformation of models that can be done automatically or interactively, as long as it is instrumented [36]. It also allows you to study and better explain the different concerns in the information analysis and the design of a process. The MDE approach also offers the possibility to interconnect all of these aspects while managing the different levels of abstraction. Figure 1 presents some concepts of the MDE.

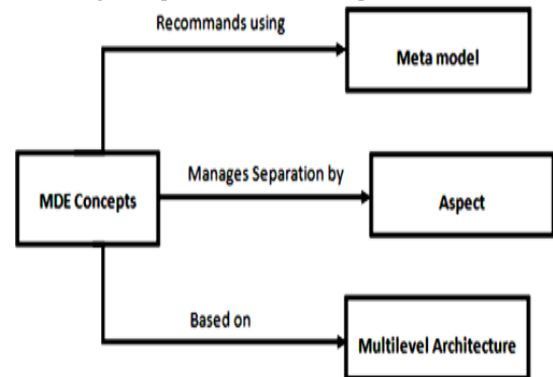


Figure 1: Some MDE concepts [37]

The construction methodology of the proposed meta model, as shown in figure 2, follows two steps:

- The first step consists of the extraction of the part that represents the process and the extension for "lifecycle", based on the "Process Structure View extended for lifecycle" proposed in the framework of the seven views approach [10]. In the meta model extracted, we eliminate the different phases that inherit from the meta-class "Phase". Then, we change the name of the meta-class "lifecycle" into "Cycle", and create two other meta-classes: "Lifecycle" and "Improvement Cycle" that inherit from the meta-class "Cycle". The output of this step is a process Structure view extended for cycles.
- The second step consists in the fusion of the process structure view extended for cycles with Concept View,

also proposed by [10]. The output of this stage is our extended process meta model extended for cycles

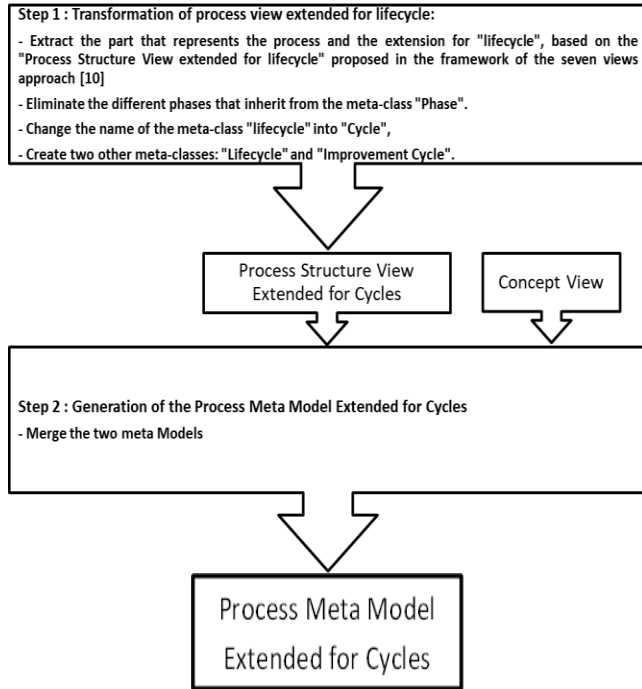


Figure 2: Methodology of generating the meta model

4 Transformation of Process Structure View Extended for Lifecycles

Based on the seven views approach [10], the meta model that we use represents the Process Structure View extended for lifecycles. This meta model allows to capture the structure and terminology of the process, and constitutes the basis for any type of correspondence between the different processes and standards,

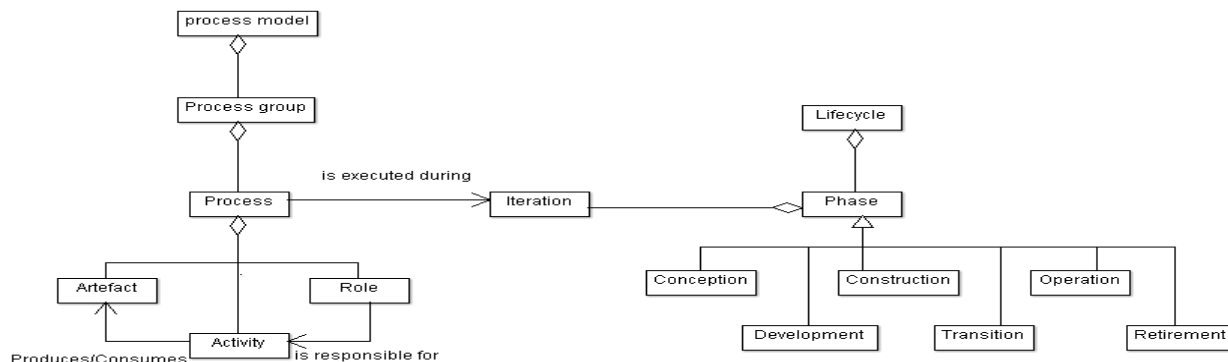


Figure 3: Process structure view extended to lifecycles [10]

which is important during the conduct of audits and evaluations. The extension of this view to lifecycles comes from the fact that it allows you to identify problems related to the initial design of the process. Figure 3 presents the process structure view extended to lifecycles.

From this meta model, we extract the part that represents the process: its deliverables, its activities and their roles, without resuming the meta-classes Process Group and Process Model. This restriction comes from the fact that we will use later the concept view meta model, which already contains these meta-classes.

Subsequently, we change the name of the meta-class "lifecycle" into "Cycle", since we propose a meta model that is valid for the implementation and improvement of the process. Of this fact, we create two meta-classes: "lifecycle" and "Improvement Cycle" which inherit the meta-class "cycle"; note that the each "lifecycle" requires an "improvement cycle"; so we will mention this specification also in the generated meta model. We will also remove the different phases indicated in the process structure view extended for lifecycles, since we are constructing a generic meta model. Each process responsible will be able to add classes that inherit the meta-class "Phase" and who represent the various phases of the cycle in question.

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Afterwards, we change the name of the meta-class "lifecycle" into "Cycle", since we propose a meta model that is valid for the implementation and improvement of the process. Of this fact, we create two meta-classes: "lifecycle" and "Improvement Cycle" which inherit the meta-class "cycle"; note that the each "lifecycle" requires an "improvement cycle"; so we will mention this specification also in the generated meta model. We will also remove the different phases indicated in the process structure view extended for lifecycles, since we are constructing a generic meta model. Each process manager will be able to add classes that inherit the meta-class "Phase" and represent the various phases of the cycle in question.

Figure 4 presents the process structure view extended to cycles

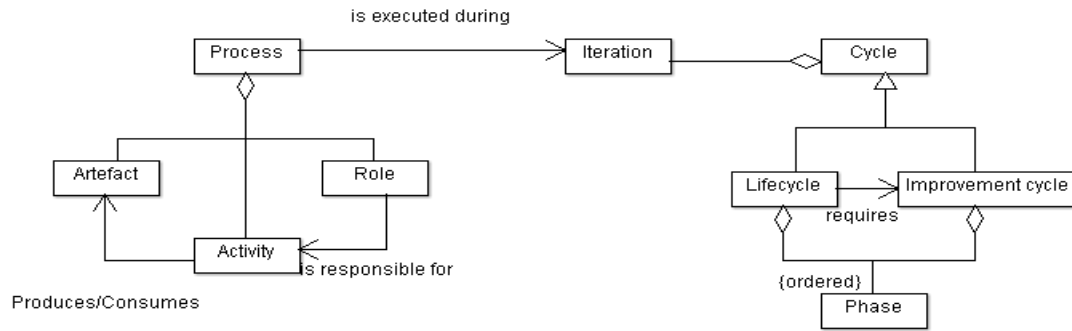


Figure 4: Process structure view extended to cycles

5 Generation of the Process Meta Model Extended for Cycles

In order to describe any process, a number of elements must be defined. The Concept View defines the elements involved in the process. This view highlights some of the problems associated with process modeling.

The class named "Process knowledge" is made up of one or more "Process". This process knowledge and its associated processes represent any sort of process knowledge. This process knowledge could be almost any sort of information relating to processes. The "Process Document" class represents the final manifestation of the process definition in some sort of document. This could be a standard, a procedure, or, indeed, some sort of web-based document. This document is formatted according to the "Document Template", which will probably reflect some in-house or corporate style of presentation of documents. This document template is made up of a number of particular sections, subsections, etc., which are simply represented on the model as the class "Section". Between the raw "Process Knowledge" and

the final deliverable process document, lies the "Process model" and it is this process model that represents an ordered, structured and consistent representation of the process knowledge. The process document is based directly on this "Process Model". Another key element of this diagram is the relationship between "Process document" and the "Requirements set" and, in particular, the numbering ratio between the two. Note that the diagram is read as: each "Requirements set" describes the purpose of one or more "Process document". It is quite often the case where different process documents, for example standards, are produced based on the same information. Bear in mind that the diagram also says: the "Process document" presents a stakeholder's view of the "Process model", which means that each stakeholder will have their own view on the process model. Therefore, it is possible for a single process model to be realized in a number of process documents that, although based on the same source information, will represent a different stakeholder's point of view [1].

The whole process meta model proposed by [10] is a generic template that is intended to be used by any organization. This meta model "Concept View" is shown in figure 5.

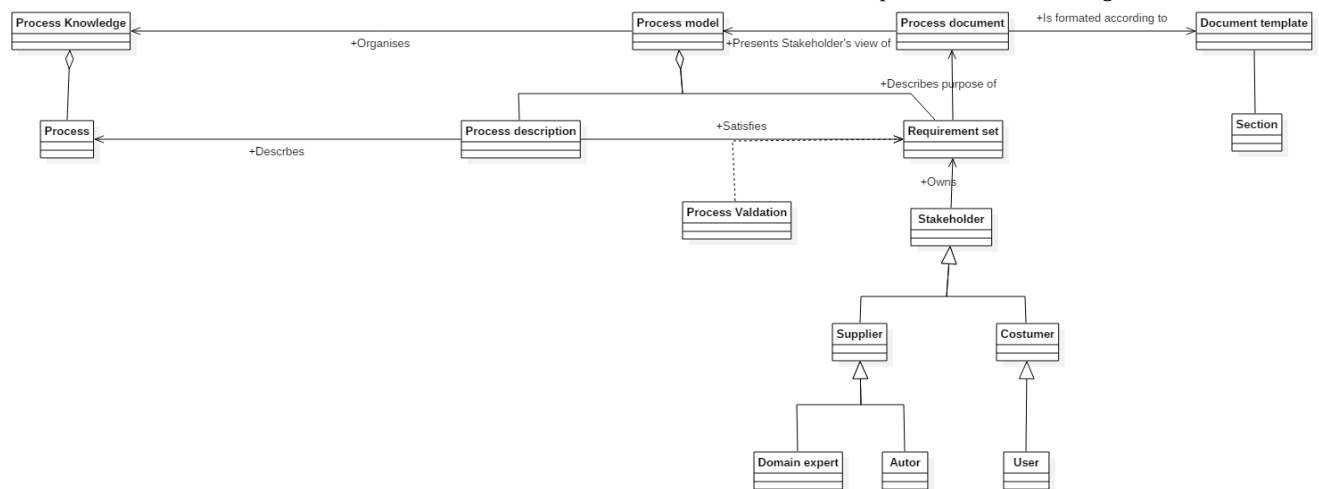


Figure 5: Concept view [10]

In this step, we merge the concept view meta model and the process structure view extended for cycles resulting from the previous step, at the level of the meta class "process", to build our

meta model "process meta model extended to cycles" presented in figure 6.

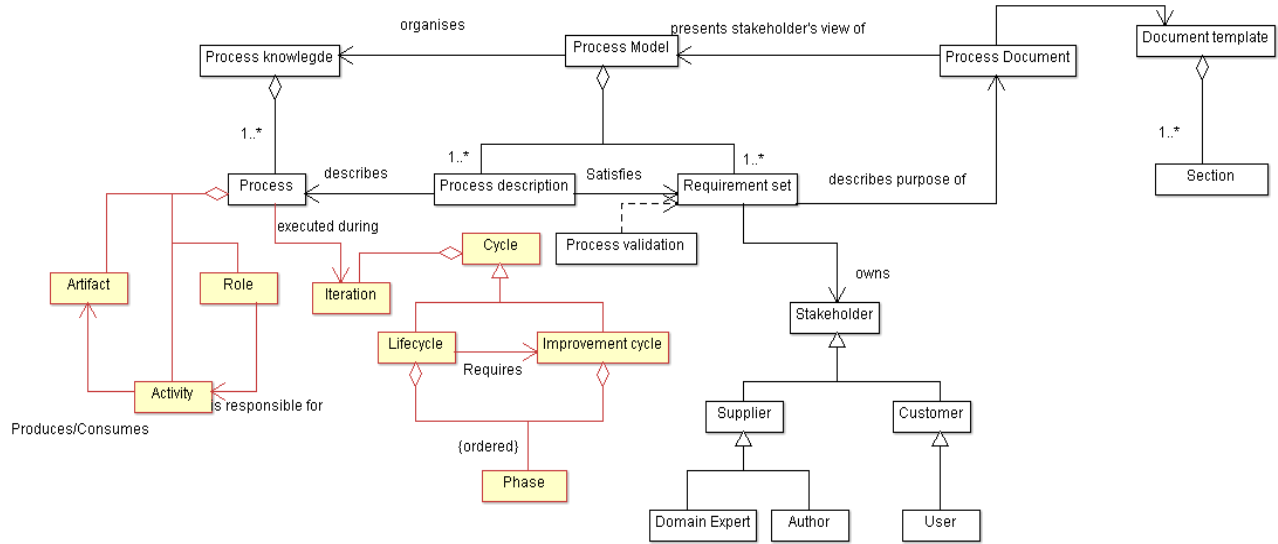


Figure 6: Our Process meta model Extended for Cycles

6 Case Study

To show the importance of the extension of the process meta model for the cycles, we will present the case of the improvement cycle PDCA. The PDCA cycle is a general process of acquisition of knowledge [38]. This process can be applied to the development of the strategic plan of a company as well as to the improvement of production series. Ishikawa has introduced in the United States and in Europe the PDCA cycle in the 60s. At this time, American consultants named it "Weel of Deeming" [39]. Deming refers explicitly in its work to the cycle of Shewhart [40].

This cycle is applied to the improvement of products and services as Shewhart had imagined. However, Deming has made, cross the theory of management, a decisive step threw introducing a new one that consists of testing the product continuously during its use [41]. It includes as well:

- designing the product (and the re-design);
- achieving;
- putting it on the market;
- testing in service.

By extension, the method can be defined as an approach based on four steps, whose main objective is the continuous

improvement of any activity and the ongoing acquisition of knowledge. The PDCA letters symbolize these steps:

- Plan: What we are going to do?
- Do: What is realized?
- Check: What is measured?
- Act: What is decided to be correct, improved?

The PDCA cycle summarizes a management philosophy that can be put in parallel with the scientific method defined in the nineteenth century by Claude Bernard. This is therefore not a "tool" to store among the traditional tools for the improvement of the quality, but the symbol of any experimental and rational approach [39].

Plan step consist of planning a realization (for e.g. write a book of specifications, establish a schedule). Step Do corresponds to the construction, the realization of a product (service). Step Check aims to control by various means (indicators, etc.) that the work (DO) corresponds well to what was planned (PLAN). Finally, the step act is a search for tracks of improvement. It must bring a new project to achieve, therefore a new planning to establish. It is therefore a cycle that it often represented by a wheel [41].

Figure 7 presents a meta modeling of a process with the application of the approach PDCA as cycle of improvement:

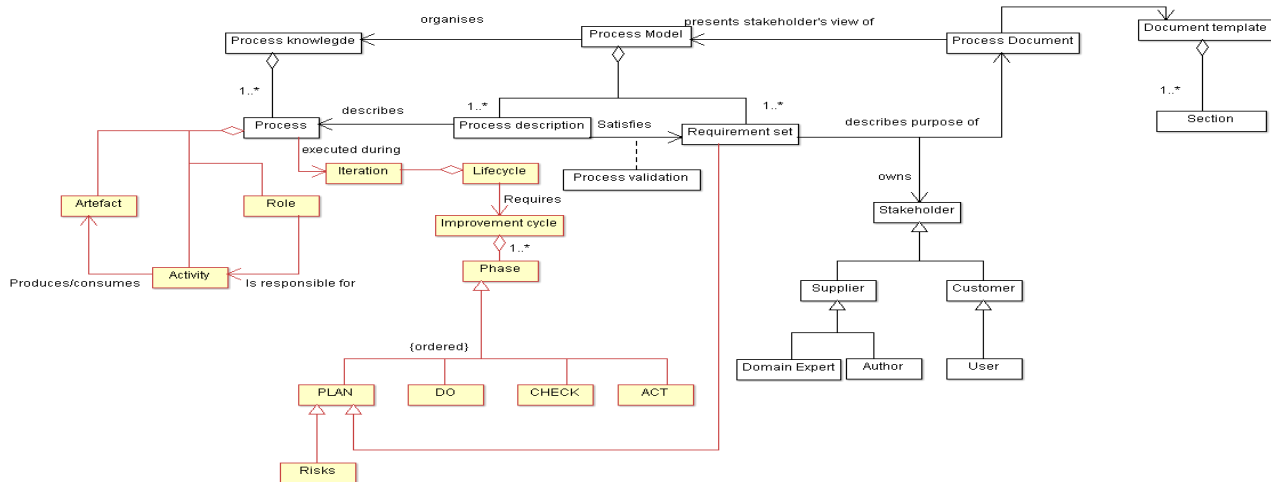


Figure 7: Our process meta model with PDCA as Improvement Cycle

In this example, we are particularly interested in the phase plan of the PDCA cycle.

The phase plan is the initial state that leads to the preparation of a test of a comparative test or experience. The complete cycle is based on the first step. The precipitated start of a project is likely to cause unnecessary costs and frustrations. We have found that the responsible for continuous improvement tend to shorten this step, to move rapidly to the second, in order to not to be inactive [42].

In this step, we must anticipate, plan and prepare. The project team captures the risks, activities, resources, means, objectives of management, assessment devices [43]. The collection of requirements and/or the formalization of new ones are to be the most essential tasks.

In the phase plan, the identification of requirements remains the more important step. In this sense, we offer through our meta model, thanks to the view requirements of the seven views approach, which is linked, to the improvement cycle. This will ensure the traceability and the good management of requirements and their impacts.

The meta model proposed will store the information collected in repositories. A repository will contain all the requirements formulated and the tracks for improvement identified as early as the stage of theoretical modeling of each process.

When the theoretical modeling of the process is not part of a comprehensive approach that includes the cycle of improvement, there would have been a redundancy during the formulation of the requirements in the course of the phase plan. In fact, the responsible for the process will be facing a problem of requirements traceability. This lack of a global repository will generate a problem of traceability of process data, for instance, the requirements from the phase plan and the avenues for improvement resulting from the phase act.

Figure 8 shows a diagram, which explains the process of adding and updating the repositories:

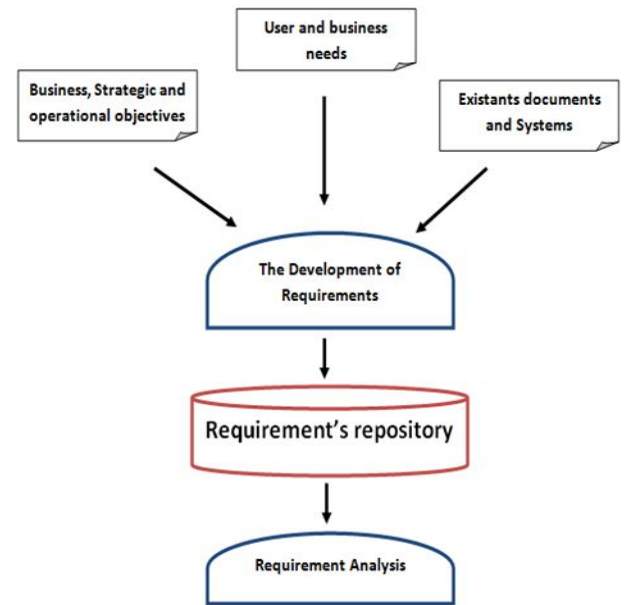


Figure 8: Process of adding information and updating repositories

While improving an existing process, during the Plan phase, and thanks to the repository already exists (traceability guaranteed), we analyze first the requirements to identify those for which the process does not respond, and develop new requirements if the case requires it.

The goal is to enrich the repository continuously.

The meta model proposed will supply the repository during the implementation phase and enrich it during the process improvement cycle.

7 Conclusion

The process modeling is an essential mandatory step in any project, either in the implementation or in the improvement phase.

However, it presents a binding and problematic step. In fact, there are several standards of modeling, each one presents specific characters for each type and area of process, and this diversity makes the choice between different models very difficult.

This article proposes a generic process meta model, extended for cycles, through the transformation and the fusion of two meta models: the concept view and the Process Structure view extended for Lifecycles, using the Seven Views approach and the concepts of the Model Driven Engineering (MDE).

The meta model resulting presents a process meta model extended for cycles (PMMEC). This meta model will allow the process responsible to keep the traceability of requirements, risks and tracks for improvement collected during the theoretical modeling, using the repositories created to store information related to the process, and which are updated at each improvement cycle.

We propose, as prospects for this work, an SoS package linked to this, which will ensure the System view in the context of complex systems.

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